

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY****First Semester of B. Tech Theory Examination (CL/ME/EE)****April – May 2018****CL 141 ENGINEERING MECHANICS****Date: 11.05.2018, Friday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. **Section I and II must be attempted in separate answer sheets.**
3. Make suitable assumptions and draw neat sketch wherever required.
4. **Use of scientific calculator is allowed.**

**SECTION – I****Q - 1 Do as directed. (Attempt any seven)****[07]**

- i) What do you mean by axis of symmetry?
- ii) The unit of area is \_\_\_\_\_.  
(a)  $\text{mm}^2$                       (b) Pa                      (c) mm
- iii) Give the name of any one scalar quantity.
- iv) What is rigid body?
- v)  $1\text{kgf} = \underline{\hspace{2cm}} \text{ N}$   
(a) 10                      (b) 5                      (c) 8
- vi) State Newton's first law.
- vii) Centroid of the line of length L is located at \_\_\_\_\_.  
(a)  $L/2$                       (b)  $2L$                       (c)  $3L$
- viii) Define magnetic force.

**Q - 2 (a) Answer the following.**

- i) Give two ways of increasing and reducing friction. **[03]**
- ii) A human's center of gravity can change as he takes on different positions. True or false? Justify your answer. **[03]**
- iii) Define the term Kinematics. **[01]**

**Q - 2 (b) i) Define term Energy and explain types of mechanical energy.****[03]**

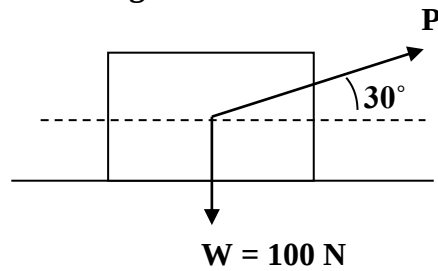
ii) Explain D'Alembert's Principle.

**[04]****OR**

**Q - 2 (b)** You may have such an experience inside an elevator: When the elevator accelerates upwards, you weigh more than you actually do, and when the elevator decelerates, you weigh less. Explain this phenomenon with an example. **[07]**

**Q - 3 (a)** A body of weight 100 N rests on a rough horizontal plane and is acted upon by a force applied at an angle of  $30^\circ$  to the horizontal. Determine the force required to just cause **[07]**

the body to slide over the surface in the direction of applied force. Take coefficient of static friction is 0.3. Refer **Figure 1**.



**Figure 1 [Q-3 (a)]**

**OR**

**Q - 3 (a)** i) Explain with the help of a neat diagram, the concept of limiting friction. **[03]**

ii) Friction is necessary evil. Comment on the validity of the statement. **[04]**

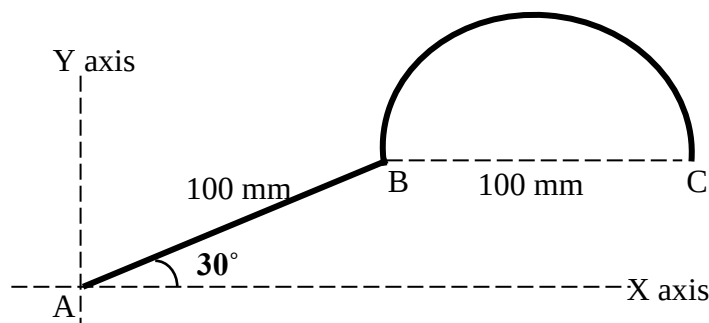
**Q - 3 (b)** i) Differentiate between center of gravity and centroid. **[03]**

ii) Give the position of centroid of the following standard sections: **[04]**

Rectangle, Triangle, Solid circle, Semicircle

**OR**

**Q - 3 (b)** Find the centroid of the bent weir shown in **Figure 2**. Consider length of AB is 100 mm **[07]**  
and diameter of semicircle BC is 100 mm.



**Figure 2 [Q-3 (b) OR]**

## **SECTION - II**

**Q - 4** **Do as directed.** **[07]**

i)  $\text{N/mm}^2$  is the unit of \_\_\_\_\_.

(a) Stress                      (b) Force                      (c) Weight

ii) A \_\_\_\_\_ quantity is one that has magnitude as well as direction.

(a) Vector                      (b) Volume                      (c) Area

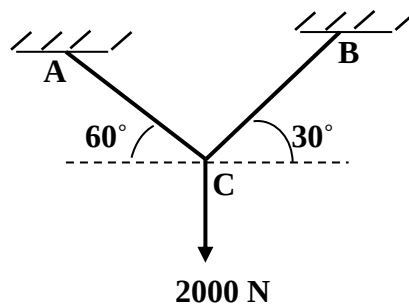
iii) Enumerate the characteristics of a force.

- iv) A body which deforms under the action of applied forces is called the \_\_\_\_\_.  
 (a) space (b) deformable body (c) static
- v) State Newton's third law.
- vi) "If three forces acting at a point are in equilibrium, then each force will be proportional to the sine of angle between other two forces". This is the statement of \_\_\_\_\_.  
 (a) Parallelogram law of forces (b) Lami's theorem (c) Varignon's theorem
- vii) Law of moments is also called the Varignon's theorem. True or false?

**Q - 5 Attempt any three.**

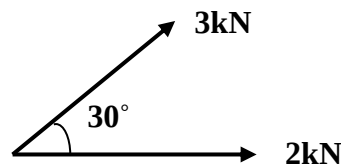
**[15]**

- (a) A weight of 2000 N is suspended from rope AC and BC as shown in **Figure 3**. Calculate the tension in each rope.



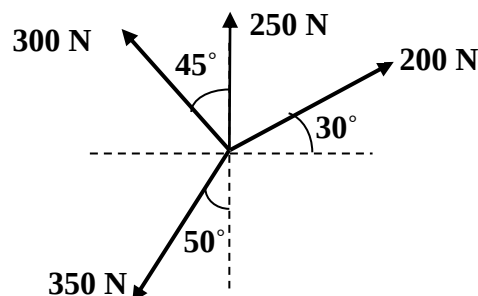
**Figure 3 [Q-5 (a)]**

- (b) Two forces of magnitude 2 kN and 3 kN are acting at a point as shown in **Figure 4**. Calculate the resultant of these concurrent force system.



**Figure 4 [Q-5 (b)]**

- (c) State and explain Parallelogram law of forces.
- (d) Calculate the resultant of coplanar concurrent force system shown in **Figure 5**.

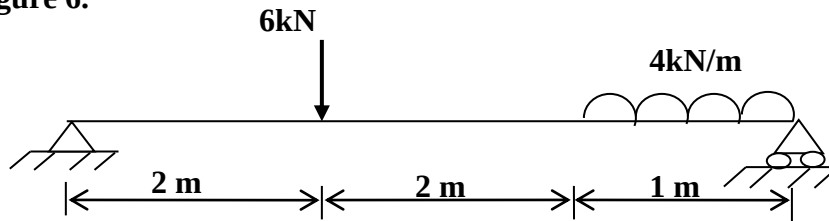


**Figure 5 [Q-5 (d)]**

**Q - 6 (a)** Enlist and explain types of load with neat sketch. [02]

**Q - 6 (b)** Calculate support reaction for the simply supported beam of span 5 m shown in [05]

**Figure 6.**



**Figure 6 [Q-6 (b)]**

**OR**

**Q - 6 (b)** Explain the steps for calculating beam reactions with neat sketch. [05]

**Q - 6 (c)** Attempt any two. [06]

- i) Enlist the types of force system and explain any one in detail.
- ii) What is couple? Enumerate its various characteristics.
- iii) Enlist different types of force.

\*\*\*\*